

SECTION A

Note: Attempt ANY FIVE Questions.

(5x4 =20 marks)

- I. Find the Reduced Grammar equivalent to the grammar G whose productions are:
 $S \rightarrow AB|CA, \quad B \rightarrow BC|AB, \quad A \rightarrow a, \quad C \rightarrow aB|b.$
- II. Design a Push Down Automaton that accepts the language $L = \{a^n c a^n | n \geq 1\}$ by final state.
- III. State Pumping lemma for regular languages. Use it to prove that the language $L = \{a^p | p \text{ is prime}\}$ is not regular.
- IV. Design a Turing Machine that accepts the language $L = \{a^n b^n | n \geq 1\}$
- V. Show that complement of a recursive language is also recursive.
- VI. Design a Turing machine to compute the sum of two positive integers n and m , where $n, m \geq 1$.

SECTION-B

Note: Attempt ALL Questions.

(5 x3 =15 marks)

- I. Convert the following Context Free grammar to equivalent CNF.

$S \rightarrow aAbB \quad A \rightarrow Ab|b \quad B \rightarrow Ba|a$

- II. Convert the following Context Free Grammar to equivalent GNF.
 $S \rightarrow YY|0 \quad Y \rightarrow SS|1$
- III. Write a short note on Universal Turing Machine (UTM).
- IV. Describe halting problem of Turing machine.
- V. Discuss Church Turing Thesis in detail. Why is it not called a theorem? Explain.

SECTION C

Note: Attempt ANY THREE Questions.

(3x5=15 marks)

- I. Q.1 If a language L and its complement $\bar{L}$ are both recursively enumerable then prove that L is recursive.
- II. Given a Context Free Grammar $G=(\{S,A,B\},\{0,1\},P,S)$ with the production set as follows:
 $S \rightarrow 0B|1A \quad A \rightarrow 0S|1AA|0 \quad B \rightarrow 1S|0BB|1$
 Design a PDA M corresponding to this CFG.
- III. Design a Turing Machine to accept the language $L=\{WW^R \mid W \in \{a,b\}^*\}$
- IV. Discuss any five variants of the Turing machine with proper model diagram and transition function.

Course Outcome

- CO1-Construct FA and minimize automata
- CO2-Construct an automaton for a given regular expression
- CO3-Construct a pushdown automaton for a given context-free language
- CO4-Construct a Turing machine deciding a given problem
- CO5-Prove whether a language is or is not a regular or context-free by using the Pumping Lemma
- CO6- Prove that a given context-free grammar generates a given context-free language
- CO7- Prove un-decidability of a given problem by reducing from a known un-decidable problem

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University Roll No.

End Term Examination, Odd Semester 2022-23
B.Tech. (CSE/AIML/CCV/DA/CSF/IIOT), III Year, V Sem
BCSC0011 THEORY OF AUTOMATA & FORMAL LANGUAGES
Maximum Marks: 50

Time: 3 Hours

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Design a PDA that accepts the language $L = \{a^n c a^n n \geq 1\}$ by final state.	4	CO3	A	P
2	State Pumping Lemma for regular languages. Use it to prove that the language $L = \{a^m m \text{ is prime}\}$ is not regular.	4	CO5	R	F
3	Design finite automaton corresponding to the given regular expression: $(a+b)^*cd^*e$	4	CO1	A	P
4	Design a Turing Machine to reverse the binary string on the input tape.	4	CO4	A	P
5	What is Universal Turing Machine? How it led to the concept of general-purpose computing machine? Or Convert the following CFG to equivalent CNF: $S \rightarrow aAbB, A \rightarrow Ab b B \rightarrow Ba a$	4	CO4	R	F

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Show that the following grammar is ambiguous. $S \rightarrow a Sa bSS SSb SbS$	3	CO3	U	C

7	Write a short note on Church Thesis. Why is it a thesis not a theorem?	3	CO7	R	
8	Design a Turing Machine to add two positive integers.	3	CO4	A	P
9	Define halting problem of Turing Machine.	3	CO4	R	F
10	Convert the following CFG to equivalent GNF: $S \rightarrow YY 0$ $Y \rightarrow SS 1$	3	CO3	An	P

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Discuss any five variants of the Turing machine with proper model diagram and transition function.	5	CO4	R	F
12	Construct a PDA accepting $\{a^n b^m a^n \mid m, n \geq 1\}$ by null store. Construct the corresponding CFG accepting the same set.	5	CO3	A	P
13	If a language L and its complement L' are both recursively enumerable then prove that L is recursive.	5	CO7	An	C

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

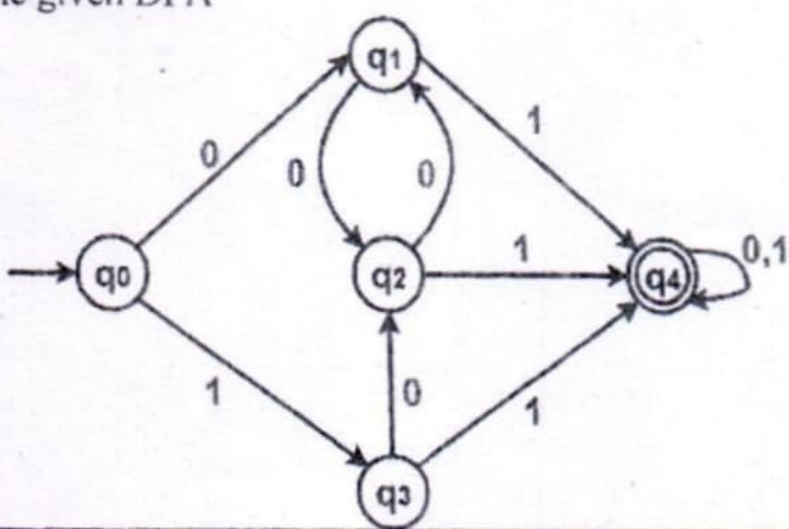
- CO1- Understand the basic concepts of Context Free languages, Expression and Grammars.
 CO2- Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.
 CO3- Analyze the process to convert regular expression to DFA, DFA to regular expression, and minimization of DFA.
 CO4- Develop the PDA for the context free language and context free grammar.
 CO5- Analyze that the grammar is ambiguous or unambiguous.
 CO6- Apply the process to convert CFG to CNF and GNF.
 CO7- Understand the concept of Turing machine and its variants.
 CO8- Design the Turing machine for the real world application.

End Term Examination, Even Semester 2022-23**B. Tech (CSE), III Year, VI Semester****Theory of Automata & Formal Languages (BCSC0011)****Time: 3 Hours****Maximum Marks: 50**

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section - A**Attempt All Questions****4 X 5 = 20 Marks**

No	Detail of Question	Marks	CO	BL	KL
1	Execute all the steps of minimization to minimize the given DFA 	4	3	An	P
2	State and prove Pumping lemma for Regular languages. Demonstrate how the set $L = \{a^n b^n \mid n \geq 0\}$ is not Regular.	4	3	R	F
3	Analyze and Design a PDA for $L = \{w \mid n_a(w) = n_b(w)\}$. Also execute the designed PDA on string abba.	4	4	C	P

4	Eliminate the useless productions from the given grammar. Enlist all the steps clearly. $S \rightarrow aaB \mid abA \mid aaS$ $A \rightarrow aA$ $B \rightarrow ab \mid b$ $C \rightarrow ad$	4	6	A	C
	OR Eliminate null productions from the given grammar. Enlist all the steps clearly. $S \rightarrow ABAC$ $A \rightarrow aA / \epsilon$ $B \rightarrow bB / \epsilon$ $C \rightarrow d$				
5	Demonstrate and Prove that complement of a recursive language (L') is also recursive.	4	7	R	P

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the following productions: $S \rightarrow aB \mid bA$ $A \rightarrow aS \mid bAA \mid a$ $B \rightarrow bS \mid aBB \mid b$ For the string aaabbabbba Generate: i) Leftmost Derivation ii) Rightmost Derivation iii) Parse Tree or Derivation Tree	3	5	U	P
7	Convert the following CFG into CNF $S \rightarrow aAD, A \rightarrow aB \mid bAB, B \rightarrow b, D \rightarrow d$	3	6	R	C
8	Analyze and Construct a Turing Machine for $L = \{WW^R \mid W \in (0,1)^*\}$	3	8	C	P
9	Explore the importance of Church Turing Thesis. Why is it not called a theorem? Explain.	3	7	R	F
10	Find a PDA equivalent to the given grammar $S \rightarrow aSbb \mid a$ Simulate the designed PDA for the string aabb	3	1,5	A	P

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Discuss any five variants of Turing Machine with proper model diagram and transition function.	5	7	R	F
12	Construct a context free grammar G which accepts M(A), where $A = (\{q_0, q_1\}, \{a, b\}, \{Z_0, Z\}, \delta, q_0, Z_0, \phi)$ where δ is given by: $\delta(q_0, b, Z_0) = \{(q_0, ZZ_0)\}$ $\delta(q_0, b, Z) = \{(q_0, ZZ)\}$ $\delta(q_1, b, Z) = \{(q_1, \epsilon)\}$ $\delta(q_0, \epsilon, Z_0) = \{(q_0, \epsilon)\}$	5	5	A	M
13	Construct and Analyze a Turing machine to compute the given function: $f(n) = \begin{cases} n-2 & \text{if } n \geq 2 \\ 1 & \text{if } n < 2 \end{cases}$	5	8	C	P

Course Name: THEORY OF AUTOMATA & FORMAL LANGUAGES

Course Outcome

- CO1- Understand the basic concepts of Context Free languages, Expression and Grammars.
- CO2- Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.
- CO3- Analyze the process to convert regular expression to DFA, DFA to regular expression, and minimization of DFA.
- CO4- Develop the PDA for the context free language and context free grammar.
- CO5- Analyze that the grammar is ambiguous or unambiguous.
- CO6- Apply the process to convert CFG to CNF and GNF.
- CO7- Understand the concept of Turing machine and its variants.
- CO8- Design the Turing machine for the real world application.

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University Roll No.

End Term Examination, Odd Semester 2023-24

B. Tech (CSE/AIML/CCV/DA/CSF/IIoT), III Year, V Sem Theory of Automata & Formal Languages (BCSC0011)

Time: 3 Hours

Maximum Marks: 50

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL																								
1	Design a Mealy Machine that can simulate as a binary Adder.	4	2	C	P																								
2	Construct a deterministic finite automata equivalent to the regular expression: $(a + b)^*(aa + bb)(a + b)^*$	4	3	E	P																								
3	Minimize the given automata: <table border="1"><thead><tr><th>State/ Inputs</th><th>0</th><th>1</th></tr></thead><tbody><tr><td>→ q₀</td><td>q₁</td><td>q₀</td></tr><tr><td>q₁</td><td>q₀</td><td>q₂</td></tr><tr><td>q₂</td><td>q₃</td><td>q₁</td></tr><tr><td>q₃</td><td>q₃</td><td>q₀</td></tr><tr><td>q₄</td><td>q₃</td><td>q₅</td></tr><tr><td>q₅</td><td>q₆</td><td>q₄</td></tr><tr><td>q₆</td><td>q₅</td><td>q₆</td></tr></tbody></table>	State/ Inputs	0	1	→ q ₀	q ₁	q ₀	q ₁	q ₀	q ₂	q ₂	q ₃	q ₁	q ₃	q ₃	q ₀	q ₄	q ₃	q ₅	q ₅	q ₆	q ₄	q ₆	q ₅	q ₆	4	3	E	P
State/ Inputs	0	1																											
→ q ₀	q ₁	q ₀																											
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q ₅	q ₆	q ₄																											
q ₆	q ₅	q ₆																											
4	Design a Turing machine for: $f(n) = \begin{cases} n-2 & \text{if } n \geq 2 \\ 1 & \text{if } n < 2 \end{cases}$	4	8	R	F																								
5	Suppose G be the grammar given as: $S \rightarrow 0B \mid 1A, A \rightarrow 0 \mid 0S \mid 1AA, B \rightarrow 1 \mid 1S \mid 0BB.$ For the string 00110101 find: a) LMD b) RMD c) Derivation Tree	4	1	E	P																								

Section – B

Attempt All Questions

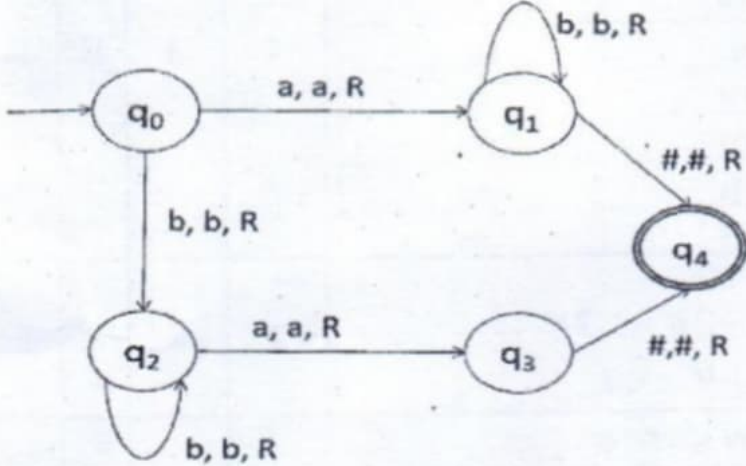
3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Find a PDA equivalent to the grammar $S \rightarrow aSbb \mid a$. Simulate the designed PDA on string aabb .	3	4	E	C
7	Convert the following grammar into CNF: $S \rightarrow aAD, A \rightarrow aB \mid bAB, B \rightarrow b, D \rightarrow d$	3	6	A	P
8	Convert the following grammar into GNf: $S \rightarrow AA \mid 0$ $A \rightarrow SS \mid 1$	3	6	A	P
9	Design a TM to recognize the strings $(ab)^+$.	3	8	C	P
10	Simplify the following grammar : $S \rightarrow 0A0 \mid 1B1 \mid BB, A \rightarrow C, B \rightarrow S \mid A, C \rightarrow S \mid \epsilon$ OR Write short note on Church Thesis. Why it is a thesis not a theorem? Explain.	3	7	R	F

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Consider the following Turing Machine M. The reject state q_{rej} is not shown, and all “missing” transitions are assumed to go to the reject state. # is the blank symbol.  a. Execute the transition function $\delta(q_2, a)$ b. Describe the computation of M on input ab. c. What is the language recognized by M?	5	8	An	P

12	Design a Turing Machine for $L = \{ww^R \mid w \in (a, b)^*\}$ and simulate the designed machine on string abbaabba .	5	8	C	P.
13	Discuss any five variants of Turing Machine with proper model diagram and transition function. OR If a language L and its complement L' are both recursively enumerable then prove that L is recursive.	5	7	U	C

CO1-Understand the basic concepts of Context Free languages, Expression and Grammars.

CO2-Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.

CO3-Analyze the process to convert regular expression to DFA, DFA to regular expression, and minimization of DFA.

CO4-Develop the PDA for the context free language and context free grammar.

CO5-Analyze that the grammar is ambiguous or unambiguous.

CO6-Apply the process to convert CFG to CNF and GNF.

CO7-Understand the concept of Turing machine and its variants.

CO8-Design the Turing machine for the real world application.

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University Roll No.

End Term Examination, Even Semester 2023-24

B. Tech (CSE), III Year, VI Sem

Theory of Automata & Formal Languages (BCSC0011)

Time: 3 Hours

Maximum Marks: 50

Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Suppose a leading firm in India has a unique password setting scheme. Their employees must create their password adhering to the following rules: i) It should start with the letters {i,j,k} ii) The fifth symbol from left end should be i. Create an FSA for this password scheme. Give the five-tuples structure for this FSA.	4	2	C	P
2	Describe the following sets by regular expressions: i. The set of strings of a's and b's ending with ab. ii. The set of strings of x's and y's and beginning with x and ending with y. iii. Find the regular expression from the given DFA	4	3	E	P

3	Convert the following DFA in Regular Expression using Arden's Theorem:	4	3	E	P
4	Find simplified and reduced grammar equivalent to the grammar G, having production rules, P: $S \rightarrow AC \mid B$, $A \rightarrow 0$, $C \rightarrow 1 \mid BC$, $E \rightarrow 0A \mid 1$.	4	8	R	F
5	Suppose G be the grammar given as: $S \rightarrow aB \mid bA$, $A \rightarrow a \mid aS \mid bAA$, $B \rightarrow b \mid bS \mid aBB$. For the string bbaababa find : a) LMD b) RMD c) Derivation Tree	4	1	E	P

Section - B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	If a language L and its complement L' are both recursively enumerable then prove that L is recursive.	3	4	E	C
7	Convert the following grammar into CNF: $S \rightarrow ASA \mid bA$, $A \rightarrow B \mid S$, $B \rightarrow c$	3	6	A	P
8	Convert the following grammar into GNF: $S \rightarrow AA \mid 0$ $A \rightarrow SS \mid 1$	3	6	A	P
9	Write short note on Church Thesis. Why it is a thesis not a theorem? Explain.	3	8	C	P
10	Find a PDA equivalent to the grammar $S \rightarrow 0S11 \mid 0$. Simulate the designed PDA on string 0011	3	7	R	F

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Details of Question	Marks	CO	BL	KL
11	Consider the following Turing Machine M. The reject state q_{rej} is not shown, and all “missing” transitions are assumed to go to the reject state. # is the blank symbol. <pre> graph LR start(()) --> q0((q0)) q0 -- "a, a, R" --> q1((q1)) q0 -- "b, b, R" --> q2((q2)) q1 -- "b, b, R" --> q1 q1 -- "#, #, R" --> q4(((q4))) q2 -- "b, b, R" --> q2 q2 -- "a, a, R" --> q3((q3)) q3 -- "#, #, R" --> q4 style start fill:none,stroke:none style q4 fill:none,stroke:2 </pre> a. Execute the transition function $\delta(q_0, a)$? b. Describe the computation of M on input bba. c. Write all tuples of turing machine	5	8	An	P
12	Design a Turing Machine for $L = \{a^n b^n c^n a, b, c \text{ are inputs and } n \geq 1\}$ and simulate the designed machine on string aabbcc.	5	8	C	P
13	Discuss any five variants of Turing Machine with proper model diagram and transition function.	5	7	U	C

BCSC 0011 Theory of Automaton and Formal LanguagesCourse Outcome

- CO1: Understand the basic concepts of Context Free languages, Expression and Grammars.
 CO2: Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.
 CO3: Analyze the process to convert regular expression to DFA, DFA to regular expression, and minimization of DFA.
 CO4: Develop the PDA for the context free language and context free grammar.
 CO5: Analyze that the grammar is ambiguous or unambiguous.
 CO6: Apply the process to convert CFG to CNF and GNF.
 CO7: Understand the concept of Turing machine and its variants.
 CO8: Design the Turing machine for the real-world application.

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University Roll No.

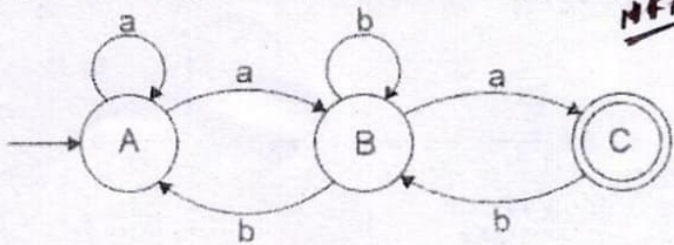
End Term Examination, Odd Semester 2024-25**B.Tech CSE, IIIrd Year, Vth Sem****BCSC 0011 Theory of Automata and Formal Languages**

Time: 3 Hours

Maximum Marks: 50

Section – A*Attempt All Questions*

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Design a DFA for the input alphabet 0,1 in which the string must be begins with 0 and ends with 1.	4	CO2	A	C
2	Find out the regular expression for the given DFA NFA 	4	CO3	A	P
3	Construct the PDA for the language having equal number of a and b over input alphabet (a, b).	4	CO4	C	C
4	What is a Recursive Enumerable Language? Prove that if L is recursive language, then the complement of L i.e. L' is also recursive.	4	CO5	R	F
5	Imagine you are part of a team designing advanced computational systems for different real-world applications. Each application requires a unique kind of computational model to address its specific constraints and goals. Your task is to understand and choose from various Turing Machine variants to solve the following challenges:	4	CO7	An	C

A. Parallel Data Processing				
B. Simultaneous Data Storage				
C. Exploring Multiple Solution				
D. Simulating General Purpose Machine				

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Explain the Church-Turing Hypothesis, provide reasoning and justification for why the Church-Turing Hypothesis is not considered as a theorem?	3	CO5	A	P
7	For the given CFG: . $S \rightarrow 0S \mid 1AA$ $A \rightarrow 0 \mid 1A \mid 0B$ $B \rightarrow 1 \mid 0BB$ For the input string $w = 0100110$ A. Construct the left most derivation B. Construct the right most derivation C. Derivation tree	3	CO5	E	F
8	Convert the given CFG into their corresponding CNF $E \rightarrow E + E \mid E * E \mid a \mid b \mid c$	3	CO6	E	P
9	Construct the PDA from the given CFG. $S \rightarrow 0SX \mid 1SY \mid \epsilon$ $X \rightarrow 1$ $Y \rightarrow 0$	3	CO4	C	P
10	You are part of a team working on a text-processing engine for a large-scale document editor. The engine must efficiently check and process structured text, such as HTML, programming code, or formulas, by using predefined grammars. To ensure that the engine operates efficiently, you are tasked with simplifying the given context-free grammar (CFG) used for parsing. CFG: $S \rightarrow aAa, A \rightarrow Sb \mid bCC \mid DaA, C \rightarrow abb \mid DD,$ $D \rightarrow aDA, E \rightarrow aC$	3	CO6	E	C

Section - C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Convert the given PDA A, into their CFG: $A = (\{q_0, q_1\}, \{a, b\}, \{a, Z_0\}, \delta, q_0, Z_0, \{\})$ $\delta(q_0, a, Z_0) = (q_0, aZ_0)$ $\delta(q_0, a, a) = (q_0, aa)$ $\delta(q_0, b, a) = (q_1, a)$ $\delta(q_1, b, a) = (q_1, a)$ $\delta(q_1, a, a) = (q_1, ^)$ $\delta(q_1, ^, Z_0) = (q_1, ^)$	5	CO4	A	P
12	Convert the grammar into their equivalent GNF: $S \rightarrow AA \mid a$ $A \rightarrow SS \mid b$	5	CO6	E	C
13	You are working on developing a data verification system for a warehouse inventory application. The system processes inventory data strings, which are structured as follows: 1. The first half of the string represents the incoming shipment (items represented by a and b). 2. The second half of the string represents the outgoing shipment, which must be reverse to the incoming shipment. Your task is to design a Turing Machine that can process these strings and determine if they are valid. If the incoming and outgoing shipments is reverse (i.e., the string conforms to the format ww^R), the Turing Machine should accept the string. Otherwise, it should reject it.	5	CO8	C	C

Theory of Automata and Formal Languages (BCSC 0011)

Course Outcome

- CO1-Construct FA and minimize automata
 CO2-Construct an automaton for a given regular expression
 CO3-Construct a pushdown automaton for a given context-free language
 CO4-Construct a Turing machine deciding a given problem
 CO5-Prove whether a language is or is not a regular or context-free by using the Pumping Lemma
 CO6-Prove that a given context-free grammar generates a given context-free language
 CO7-Prove un-decidability of a given problem by reducing from a known un-decidable problem

Printed Pages:

University Roll No.

End Term Examination, Even Semester 2024-25

B.Tech (CSE/AIIML/CCV/DA/IIoT/CSF), IIIrd Year, VIth Sem

Theory of Automata and Formal Languages (BCSC 0011)

Time: 3 Hours

Maximum Marks: 50

Instruction for students:

1. All parts of the question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Construct a DFA for the input alphabet $\{0, 1\}$ in which the binary number must be divisible by 3. OR State and Proof Arden's Theorem. Write the steps to convert any given NFA into their corresponding DFA.	4	1	C	P
2	Design a DFA for the given Regular expression (R). $R = (ab)^*(ab + ba)$	4	2	An	P
3	Construct the PDA from the given CFG: $G = (\{S, X\}, \{a, b\}, P, S)$ where the productions are: $S \rightarrow XS \mid \wedge, \quad A \rightarrow aXb \mid Ab \mid ab$	4	3	C	C
4	Simplify the given CFG (G). $G = (\{S, A, B, C\}, \{a, b\}, P, S)$ $P: S \rightarrow aAa, \quad A \rightarrow bBB, \quad B \rightarrow ab, \quad C \rightarrow aB$	4	4	C	C

5	Suppose you are designing a system that verifies user codes. The codes are composed of two characters, 'a' and 'b'. The system accepts equal number of both elements i.e. $a^n b^n$ where $n > 0$. To automate the verification process, you decide to implement a Turing Machine (TM) that checks whether a given input string is valid or not.	4	6	E	P
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Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Eliminates the null and unit production of the given CFG (G): $S \rightarrow AbaC, \quad A \rightarrow BC, \quad B \rightarrow bB \mid ^,$ $C \rightarrow D \mid ^, \quad D \rightarrow d$	3	6	E	C
7	Convert the given CFG (G) into their corresponding Chomsky Normal Form: $G: S \rightarrow Aba, \quad A \rightarrow aab, \quad B \rightarrow AC$	3	6	U	P
8	Define recursive and recursive enumerable languages.	3	7	R	F
9	Find out the LMD, RMD and Derivation Tree for the given CFG (G), for the input string $aabbaa$ $G: S \rightarrow aAS \mid SS \mid a, \quad A \rightarrow ba \mid SbA$	3	6	A	P
10	You are attending a seminar on the foundations of theoretical computer science. During a discussion on computability, a student asks why the Church-Turing Thesis is so widely accepted if it hasn't been formally proven. The speaker responds by saying, "That's because it's a thesis, not a theorem." Elaborate it.	3	6	U	C

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Construct the equivalent CFG (G) for the given PDA (A). $A = (\{q_0, q_1\}, \{0, 1\}, \{X, Z_0\}, \delta, q_0, Z_0, \{ \})$ $\delta(q_0, 0, Z_0) = (q_0, XZ_0)$ $\delta(q_0, 1, X) = (q_1, a)$ $\delta(q_1, 1, X) = (q_1, ^)$	5	6	E	P

12	Convert the given grammar CFG (G) into their corresponding GNF: G: $S \rightarrow AA \mid 0$, $A \rightarrow SS \mid 1$	5	6	A	P
13	Suppose we need to determine in how many different ways a Turing Machine can be utilized based on its variants. Explain any five specific types of Turing Machines, highlighting their unique characteristics.	5	4	R	F